

Sandboxer: An Experimental Model-versus-Model Benchmark for Offensive and Defensive Cyber Capability in a Simulated Capture-the-Flag Arena

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Abstract

SANDBOXER is an experimental benchmark that evaluates the offensive and defensive cyber capabilities of frontier AI models in a simulated, controlled, and fully observable capture-the-flag (CTF) arena. Where most existing benchmarks measure whether a single model can solve a pre-authored security challenge, SANDBOXER instead pits two models directly against one another: each model first builds and defends its own synthetic network service (Blue Phase), and then attacks the service built by its opponent (Red Phase), under symmetric, audited budgets. Vulnerabilities are never planted by the benchmark; they must emerge from the models’ own defensive choices, which is precisely what the arena is designed to reveal. A trusted Orchestrator and a hybrid deterministic-and-LLM Match Auditor enforce isolation, symmetry, and validity, while an append-only, hash-chained telemetry stream acts as the authoritative record from which results, replays, reports, and broadcast videos are derived. This paper introduces the benchmark, describes its design, protocol, scoring, evidence, and auditing machinery, and states the limitations that bound the claims it can support. It is the introductory announcement of SANDBOXER ahead of its first public best-of-three series between two open-weight models, and it is deliberately a design document, not yet a report of results.

1 Introduction

Frontier language models are increasingly deployed as autonomous agents that plan, call tools, and act on external systems over long horizons. As their autonomy grows, so does the practical importance of understanding what they can do in security-relevant settings—not only how well they can answer quiz-style questions about vulnerabilities, but how they behave when they must build a defense and, symmetrically, when they must break one under time, token, and tool constraints. Measuring these capabilities rigorously is hard: real systems cannot be attacked for evaluation purposes, and historical challenge datasets, however carefully curated, exercise only one side of the coin.

The evaluation community has responded with a family of cybersecurity benchmarks that score a model’s ability to solve curated challenges, among them Cybench [6], NYU CTF Bench [7], CyberGym [8], and the EnIGMA agent [9]. These are valuable and complementary efforts, but they share a common structure: a single agent attacks a fixed, pre-authored target, and the evaluation produces a success rate against that static corpus. They do not measure how a model defends a system it built itself, nor do they measure how two models interact adversarially under controlled, symmetric conditions.

SANDBOXER is designed to fill this gap. Its core unit is a *Match*: two models are placed in separate, disposable, isolated virtual machines. In the Blue Phase, each model receives the same minimal synthetic service and the same seeded “Blue Brief”—a small functional requirement, not

an exploit—and is free to modify and defend its own service. In the Red Phase, the two arenas are connected along narrow, declared paths, and each model attempts to capture a synthetic flag from the opponent’s service while preserving its own. The vulnerabilities that matter are therefore *emergent*: they arise from the models’ independent defensive choices, rather than being selected by the seed or pre-baked into the challenge. This makes the benchmark a test of integrated offensive and defensive behavior rather than a test of challenge-solving.

Three commitments distinguish SANDBOXER from a demonstration. First, *observability*: every state transition, tool call, budget observation, and submission is recorded in an append-only, hash-chained telemetry stream that is the authoritative source of truth; the video, the report, and the website are all derived views of that stream. Second, *symmetry*: the two competitors face identical prompts, tools, budgets, and provider treatment, and any material asymmetry invalidates the comparison rather than being papered over. Third, *bounded claims*: SANDBOXER is an experimental benchmark, and its public claim is deliberately narrow—the result describes these specific models, in this specific configuration, and does not measure general cyber capability or safety against real-world systems.

This paper is the introductory announcement of the benchmark. We first situate it against related work (Section 2), then describe the arena, the match protocol, scoring, telemetry, and auditing in detail (Section 3), and close with the limitations and threats to validity that bound what the benchmark can claim (Section 4).

2 Background and Related Work

2.1 The broader evaluation landscape

The evaluation of frontier models has matured from static question-answering benchmarks toward *capability* evaluations that ask how much useful work an agent can complete over an extended horizon. METR proposes measuring AI performance by the length of tasks an agent can complete, arguing that task-completion time horizon is a stable and interpretable metric for progress [1]. Epoch AI’s FrontierMath demonstrates the value of expert-crafted, unseen, and independently vetted problems for probing capability at the frontier [2]. The UK AI Safety Institute’s Inspect framework provides an open, extensible harness for running and logging such evaluations reproducibly [3]. In the software-engineering domain, SWE-bench [4] and InterCode [5] established the standard of scoring an agent by whether its actions, executed against a real environment, actually achieve the stated objective.

SANDBOXER inherits from this tradition the discipline of execution-based scoring, frozen configurations, and versioned evidence, but applies it to an adversarial, two-sided task rather than a single-agent completion task.

2.2 Cybersecurity benchmarks

A distinct line of work evaluates language-model agents on security tasks. Cybench provides a framework for specifying cybersecurity tasks and scoring agents on 40 professional-level CTF challenges [6]. NYU CTF Bench curates 200 CTF challenges across six categories and evaluates both black-box and open-source models [7]. CyberGym scales real-world evaluation to 1,507 historical vulnerabilities across 188 software projects [8]. EnIGMA shows that interactive agent-computer interfaces substantially improve a model’s ability to solve CTF challenges [9]. Together these efforts establish that (i) language-model agents possess non-trivial offensive capability on curated challenges, and (ii) the choice of interface, scaffolding, and budget materially changes measured performance.

What these benchmarks share is a one-sided structure: a single agent attacks a pre-authored target, and the result is a per-challenge success rate. Three dimensions are therefore under-explored. First, *defense*: there is no symmetrically rigorous measure of how well a model

constructs and maintains a defense. Second, *interaction*: real security is a contest, and a model’s behavior against an adaptive opponent is a different quantity from its behavior against a static target. Third, *emergence*: when vulnerabilities are pre-authored, the benchmark cannot distinguish the model’s skill from the difficulty of the planted flaw.

SANDBOXER is positioned at the intersection of these two lines of work. It uses the disciplined evaluation machinery of the capability-evaluation tradition (frozen configuration, execution-based scoring, versioned telemetry, an independent auditor) applied to the two-sided, emergent task structure that single-agent CTF benchmarks cannot express. It is closer in spirit to a controlled “model duel” than to a challenge corpus, and it is explicitly a *showcase evaluation*—a first, carefully bounded answer to the question of how two models compare when they both defend and attack—rather than a claim of mature scientific-benchmark status comparable to the established institutions cited above.

3 Methodology

3.1 Design principles

Four principles govern every design decision in SANDBOXER.

- **Safety first.** The arena is synthetic and self-contained. Targets are toy services and ephemeral flags; there is no path to real systems, real credentials, or the control plane. Isolation is deny-by-default and enforced by deterministic controls that fail closed.
- **Symmetry.** Both competitors receive identical prompts, tools, budgets, provider treatment, and opportunity. Any material asymmetry in a measured category invalidates the comparison unless a predeclared symmetric fallback removes that category from both sides.
- **Observability.** A single, authoritative, hash-chained telemetry stream records everything that matters. Nothing is inferred from screen recordings; the replay, the report, and the video are all derived from the same validated events.
- **Emergent vulnerability.** No exploit is planted. Both models start from the same pristine service and the same seeded functional brief; the attack surface that matters is whatever each model’s own choices open up.

3.2 Arena and containment

Each Match runs two disposable Runner virtual machines under a trusted Orchestrator. Runners are non-privileged, capability-free, and read-only at the root filesystem, with bounded CPU, memory, and process limits. During the Blue Phase their networks are fully separated; a shared, deny-by-default arena network is attached only at the transition to the Red Phase, and even then only the declared opponent service paths are reachable. There is no egress to the public internet, no host filesystem mounts, and the Orchestrator—which is never a valid target—cannot be addressed, terminated, or administered from the arena. Flags, services, and targets are synthetic; nothing outside the arena is a permitted objective. Every terminal state ends in verified teardown, and any teardown that cannot be confirmed destroys the validity of the Match.

3.3 Task family and the Blue Brief

The task is deliberately minimal so that the interesting variation comes from the models, not from an elaborate scenario. Both competitors begin from the same small, pristine, synthetic notes service with deterministic health, functional-integrity, and flag-capture tests. The service contains no planted exploit.

At the start of a Match, both competitors receive the same seed-selected **Blue Brief**: a minor functional requirement—such as constrained public export, controlled sharing, or sanitized diagnostics—stated as an outcome, not an implementation. Each model independently changes and defends its own service to satisfy the brief. The briefs are versioned, frozen for a series, and calibrated to be symmetric and of comparable difficulty. Because the brief specifies *what* to add and not *how* to secure it, the defensive choices—and therefore the attack surface—emerge from the model. A best-of-three series uses three distinct briefs selected without replacement from the frozen catalog.

3.4 Match protocol

A Match is a validated state machine controlled by the Orchestrator:

```
specified → provisioned → preflight_passed → blue_active →  
blue_validation → intermission_interview → flags_injected →  
red_preflight → red_active → finalizing → valid_result | invalid |  
aborted → teardown_verified
```

Every transition is an authoritative telemetry event with a monotonic timestamp, a reason, and a responsible component, and a failed prerequisite cannot be bypassed by advancing the state manually.

Blue Phase. Both competitors build and defend their own service in independent concurrent loops. Allowed tools are limited to inspecting, editing, restarting, and testing the model’s *own* service, plus an irreversible **finish_phase**. The phase ends when both competitors finish or exhaust their Blue budgets. The Auditor then verifies that both services pass the same minimum health contract; if either fails, the Red Phase does not start and the Match is invalid rather than an automatic competitive loss.

Intermission Interview. Between phases, each model receives the same separate, tool-free interview asking how its defensive phase went, which weaknesses it perceives, and what attack it expects. The interview uses a small equal budget, cannot change Runner state, reveals no opponent data, and is recorded and labeled as model self-report rather than verified Match fact. It cannot affect scoring except through ordinary provider failures.

Red Phase. The Orchestrator injects unique ephemeral flags, verifies their placement, and enables only the declared opponent service paths. Both competitors then run simultaneously under symmetric Red budgets. Each model may continue to defend its own service while attacking the opponent’s, and may submit a captured flag to the Orchestrator for verification. A valid submission requires the correct opponent flag *and* the submitter’s own service to pass the minimum health contract at submission time.

Budgets. The primary competitive resource is generation consumption, not elapsed time. Both competitors receive a comparable output-token allowance, equal maximum turns, equal tool availability, and equal retry and pacing policy; reasoning tokens are included symmetrically when the provider exposes them. Wall-clock limits exist only as loose operational backstops. Provider latency is observed evidence but is not itself a competitive budget, and a transient provider failure that produces no generation does not consume competitive tokens.

3.5 Scoring and the best-of-three

SANDBOXER uses an outcome-first, lexicographic result rather than an additive point total. The Orchestrator is the sole authority for flag validity, submission receipt and verification, service availability, and functional integrity. For a valid Match, the winner is determined as follows.

1. **Exactly one competitor captures:** that competitor wins, even if its own service is no longer healthy at finalization; capture is the primary objective and the degraded defense is reported as part of the result.
2. **Both capture:** if exactly one service is functionally healthy at finalization, that competitor wins; if both are healthy, the earlier verified submission wins; if neither is healthy, higher measured functional availability wins.
3. **Neither captures:** higher measured functional availability and integrity wins.
4. **Unresolved tie:** a fresh, role-swapped tie-break Match is run under the same frozen protocol, rather than a coin flip.

Time-to-capture is therefore an explicit ordering criterion only when both competitors complete equivalent healthy outcomes; it is never converted into an arbitrary point bonus. Safety and validity outcomes are applied first: an attributable disqualification awards the Match to the opponent, while an infrastructure, isolation, or authoritative-telemetry failure invalidates the Match with no winner.

A public comparison is a **best-of-three** series, first to two valid Match wins. Match 1 assigns the internal side randomly from a recorded source, Match 2 swaps it, and Match 3 (only at 1–1) assigns it to the loser of Match 2. A single Match may be published as an experimental showcase, but the best-of-three is the minimum unit called a v0 model comparison.

3.6 Telemetry, evidence, and the Auditor

Telemetry. Match Telemetry is an append-only JSONL stream with a versioned envelope. Every event carries a schema version, match and series identifiers, an event identifier, UTC wall-clock time plus Orchestrator monotonic nanoseconds, the emitting component, phase and turn, causal and idempotency identifiers, a redaction classification, the source adapter and exact software versions, and a hash chain linking each event to its predecessor. The Orchestrator is authoritative for phase transitions, budgets, network state, health, submissions, termination, validity, and score; Runner observations are untrusted until normalized and validated.

Evidence. Each evidence bundle contains the immutable Match specification, prompt and tool contracts, the seed commitment and post-Match reveal, full competitor and provider manifests, environment and image digests, raw and normalized telemetry, service diffs, health evidence, score proof, and the Auditor verdict. Secrets and raw flags are encrypted in restricted evidence or replaced by proofs; public artifacts contain only irreversible redactions. A correction creates a new artifact version and never silently rewrites history.

The Match Auditor. A trusted, independent Auditor supervises every Match. Deterministic controls are authoritative for identity, symmetry, budget categories, allowlisted tools and network destinations, telemetry continuity, submission verification, forbidden targets, credential access, privilege or sandbox-escape indicators, and control-plane reachability. An independent LLM reviewer may assess task sensibility, ambiguous novel risk, and whether interpretations are supported, but it cannot weaken a rule, fabricate a missing measurement, edit evidence, or permit a blocked action. High-confidence safety violations block and abort immediately; uncertain signals fail closed when they could expose systems outside the synthetic arena. Actions—warn, pause, abort, invalidate, disqualify, quarantine, escalate—each carry a versioned reason code and evidence references.

3.7 Competitors, provider, and preflight

The Competitor is identified by producer and exact model, and the provider is a declared boundary rather than part of Competitor identity. The first controlled series runs `poolside/laguna-s-2.1-free` (Laguna) against `meta/muse-spark-1.2-contributor` (Muse Spark Contributor), both served through the Command Code provider via a headless CLI adapter. The adapter runs the CLI non-interactively, parses its NDJSON event stream, denies all native CLI tools, and exposes only phase-scoped tools backed by the Runner bridge.

Before any Match, a **deterministic provider preflight** records the CLI binary hash and version, a redacted account fingerprint, a live catalog snapshot, exact model presence, and the accounting categories actually reported by the provider, and it live-probes the requested pair to confirm that both models resolve and are entitled. Catalog drift, a missing or aliased model, asymmetric usage categories, insufficient entitlement, or an unsupported session control fails the preflight and blocks the Match. Fields the provider does not expose (such as a credit balance or a concurrency limit) are recorded as not-observable rather than guessed, and comparability is asserted only over what is actually measured symmetrically.

3.8 Publication pipeline

All public artifacts are derived views of a frozen, validated evidence bundle, and every publication step is gated by a human approval. A Result Report separates *observed* facts, *derived* measurements, *interpreted* behavior, and *hypotheses*, and states the typing of every analytical claim. A deterministic compositor renders a split-terminal replay from sanitized telemetry, with a two-voice commentary track that is drafted, grounded to event identifiers, and reviewed; private model reasoning is redacted by construction. The report, the video, and the results website are produced from the same artifacts, and rollback never deletes evidence or silently replaces history. Maps, raw traces, flags, and credentials are never exposed publicly.

4 Limitations

SANDBOXER is deliberately narrow, and its claims must be read with the following limitations in mind.

- **Scope of the claim.** The result of a Match or series describes these specific models, in this specific configuration, on this synthetic task. It does not measure general cyber capability, safety, or performance against real-world systems, and it must not be generalized beyond the configuration actually run.
- **Synthetic task.** The toy service and the seeded briefs are a deliberate simplification. Strategies that transfer to real systems, and the full complexity of real defense, are not exercised here. The benchmark measures a controlled slice of behavior, not the whole of cybersecurity competence.
- **Small samples and variance.** A best-of-three is the minimum unit of comparison, but it remains a small sample. The methodology is not yet validated over many seeds, replications, execution orders, or model families, and no statistical significance beyond the runs performed is claimed. The v0 catalog of three to five briefs can also produce monotony; broader or more generative brief construction is a deferred, measured extension.
- **Provider and scaffold confounders.** Provider-added prompts, scaffolding, caching, rate limits, latency, and adapter behavior can all influence observed behavior and are disclosed as confounders rather than model differences. Input-token accounting is treated as evidence only, because fixed CLI context and caching are not comparable across the two competitors.

- **Token-budget divergence.** Nominal token budgets and observed consumption can diverge, and a model’s refusal, voluntary finish, or ineffective action is recorded as behavior, not as an infrastructure fault. The comparison is therefore about behavior under a declared budget, not a perfect measurement of capability.
- **Policy drift.** Provider safety and refusal policies change over time and can affect comparability; a material asymmetry or policy incompatibility is treated as a validity event rather than a result.
- **Interpretation versus observation.** Commentary, metaphor, and strategic interpretation are editorial derivatives and can misread an event. They never substitute for telemetry, and public material marks interpretation explicitly (“appears to”, “seems to”).
- **Status of the benchmark.** SANDBOXER is an experimental showcase evaluation. It does not claim mature scientific-benchmark status comparable to established evaluation institutions such as METR or Epoch AI, and EnIGMA and NYU CTF Bench are retained as methodological and editorial references, not as evidence that SANDBOXER measures the same construct.

These limitations are not defects to be hidden; they are the boundary of what the benchmark can responsibly claim. The first public series is a first step toward closing the two-sided gap in cyber evaluation, and its value is as much in the transparency of its evidence and the honesty of its claims as in any particular result it produces.

References

- [1] T. Kwa, B. West, J. Becker, *et al.*, “Measuring AI ability to complete long tasks,” arXiv preprint arXiv:2503.14499, 2025.
- [2] E. Glazer, E. Erdil, T. Besiroglu, *et al.*, “FrontierMath: A benchmark for evaluating advanced mathematical reasoning in AI,” arXiv preprint arXiv:2411.04872, 2024.
- [3] UK AI Safety Institute, “Inspect: A framework for large language model evaluations,” software framework, <https://inspect.aisi.org.uk/>.
- [4] C. E. Jimenez, J. Yang, A. Wettig, *et al.*, “SWE-bench: Can language models resolve real-world GitHub issues?,” arXiv preprint arXiv:2310.06770, 2024.
- [5] J. Yang, A. Prabhakar, K. Narasimhan, and S. Yao, “InterCode: Standardizing and benchmarking interactive coding with execution feedback,” arXiv preprint arXiv:2306.14898, 2023.
- [6] A. K. Zhang, N. Perry, R. Dulepet, *et al.*, “Cybench: A framework for evaluating cybersecurity capabilities and risk of language models,” arXiv preprint arXiv:2408.08926, 2024.
- [7] M. Shao, S. Jancheska, M. Udeshi, *et al.*, “NYU CTF Bench: A scalable open-source benchmark dataset for evaluating LLMs in offensive security,” arXiv preprint arXiv:2406.05590, 2024.
- [8] Z. Wang, *et al.*, “CyberGym: Evaluating AI agents’ real-world cybersecurity capabilities at scale,” arXiv preprint arXiv:2506.02548, 2025.
- [9] T. Abramovich, M. Udeshi, M. Shao, *et al.*, “EnIGMA: Interactive tools substantially assist LM agents in finding security vulnerabilities,” arXiv preprint arXiv:2409.16165, 2024.